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PROFESSIONAL SUMMARY

Machine Learning / AI Engineer building end-to-end ML and Retrieval-Augmented Generation (RAG) pipelines in Python - from data preprocessing, feature engineering, and model evaluation to API deployment (FastAPI) on cloud (GCP, AWS).
? Designed a full-stack RAG search engine with Sentence-Transformers embeddings, a ChromaDB vector database, and Large Language Model (LLM) integration, achieving 86.67% precision@5 on a 508-chunk retrieval benchmark.
? Built supervised machine learning classifiers for phishing URL detection and NLP sentiment/topic analysis tools processing 8,000+ videos.
? Proficient in Python, SQL, Pandas, NumPy, Scikit-learn, and LLM APIs (Anthropic Claude, Groq); pursuing a B.Tech in Computer Science (expected 2027).

EDUCATION

B.Tech in Computer Science - Graphic Era University | 2023 - 2027 (Expected)

Higher Secondary (Class XII) - The Indian Academy | 2022 - 2023

TECHNICAL SKILLS

Programming Languages: Python, Java, C, HTML, CSS, JavaScript, SQL
ML & Data Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
ML & AI Concepts: Machine Learning, Retrieval-Augmented Generation (RAG), LLM integration, embeddings, vector databases, data cleaning, preprocessing, feature engineering, model evaluation, exploratory data analysis (EDA), fraud detection, sentiment analysis, data visualization
Developer Tools: Git, GitHub, Jupyter Notebooks, FastAPI, REST API, PyMuPDF
Cloud & AI Platforms: Google Cloud Platform (GCP), Amazon Web Services (AWS), Anthropic Claude API, Groq

PROJECTS

Semantic Search & RAG Engine - Python, FastAPI, React, ChromaDB

- Engineered a full-stack Retrieval-Augmented Generation (RAG) application (FastAPI + React/Tailwind) with a ChromaDB vector database and a token-aware chunking/embedding pipeline (Sentence-Transformers).

- Built on-demand document ingestion (Wikipedia topics + PDF upload via PyMuPDF) and streaming RAG Q&A using Groq Llama 3.1 (LLM integration) across 8 REST API endpoints.

- Achieved 86.67% precision@5 on a 15-query hand-labeled retrieval benchmark spanning a 508-chunk corpus.

- github.com/yuvrajnegii/Embedding-Based-Semantic-Search

Machine Learning-Powered Phishing URL Detector

- Built a supervised machine learning classification model to detect phishing URLs using lexical and domain-based feature engineering.

- Trained and evaluated multiple ML models with Scikit-learn, achieving high precision and recall.

- github.com/yuvrajnegii/Phishtackle

YouTube Sentiment and Topic Analysis Tool

- Built an end-to-end NLP pipeline processing 8,000 videos across 4 major news channels via the YouTube Data API, applying regex-based topic detection across 6 tracked political topics.

- Identified 2,411 topic-relevant videos (30.1% of corpus) and scored each for sentiment using NLTK VADER.

- Visualized sentiment distribution and per-channel, per-topic trends with Pandas, Seaborn, and Matplotlib.

- github.com/yuvrajnegii/Sentiment-analysis

CERTIFICATES

AWS Cloud Practitioner Essentials - Amazon Web Services (AWS) | Mar 2025

Claude with the Anthropic API - Anthropic | Apr 2026